

Problem 4

Part 4a)

Let $A \in \mathcal{B}(\mathbb{R}^n)$ be an arbitrary Borel set. We want to determine the probability measure of Y :

$$\begin{aligned}\mathbb{P}(Y \in A) &= \mathbb{P}(g(X) \in A) = \mathbb{P}(X \in g^{-1}(A) \cap B) \\ &= \int_{g^{-1}(A) \cap B} f_X(x) dx.\end{aligned}$$

Since the sequence of sets $(B_i)_{i=0,\dots,m}$ forms a finite partition of B into pairwise disjoint subsets, we can decompose the integral over the disjoint regions:

$$\begin{aligned}\mathbb{P}(Y \in A) &= \sum_{k=0}^m \int_{g^{-1}(A) \cap B_k} f_X(x) dx \\ &= \int_{g^{-1}(A) \cap B_0} f_X(x) dx + \sum_{k=1}^m \int_{g^{-1}(A) \cap B_k} f_X(x) dx.\end{aligned}$$

Since B_0 is a Lebesgue null set and f_X is a density, the probability measure over this set is zero. Thus, $\int_{g^{-1}(A) \cap B_0} f_X(x) dx = 0$, so that:

$$\mathbb{P}(Y \in A) = \sum_{k=1}^m \int_{g^{-1}(A) \cap B_k} f_X(x) dx. \quad \checkmark \quad (1)$$

For each $k \in \{1, \dots, m\}$, the restriction $g_k := g|_{B_k} : B_k \rightarrow \mathbb{R}^n$ is injective, continuously differentiable, and has a regular Jacobian matrix, i.e., it satisfies the conditions for the Density Transformation Formula in $\mathbb{R}^n$ (Theorem 3.1.14). We apply the change of variables $y = g_k(x)$, which gives $x = g_k^{-1}(y)$ and $dx = |\det J_{g_k^{-1}}(y)| dy$. The domain of integration $x \in g^{-1}(A) \cap B_k$ bijectively maps to $y \in A \cap g(B_k)$. Applying this transformation to the integral yields:

$$\begin{aligned}\int_{g^{-1}(A) \cap B_k} f_X(x) dx &= \int_{A \cap g(B_k)} f_X(g_k^{-1}(y)) |\det J_{g_k^{-1}}(y)| dy \\ &\stackrel{\text{change of variables}}{=} \int_A f_X(g_k^{-1}(y)) |\det J_{g_k^{-1}}(y)| \mathbb{1}_{g(B_k)}(y) dy.\end{aligned} \quad \checkmark$$

Substituting this back into equation (1), we obtain:

$$\mathbb{P}(Y \in A) = \sum_{k=1}^m \int_A f_X(g_k^{-1}(y)) |\det J_{g_k^{-1}}(y)| \mathbb{1}_{g(B_k)}(y) dy.$$

Because the sum is finite, the linearity of the Lebesgue integral allows us to interchange the summation and integration:

$$\mathbb{P}(Y \in A) = \int_A \left(\sum_{k=1}^m f_X(g_k^{-1}(y)) |\det J_{g_k^{-1}}(y)| \mathbb{1}_{g(B_k)}(y) \right) dy. \quad \checkmark$$

Since $A \in \mathcal{B}(\mathbb{R}^n)$ was chosen arbitrarily, the integrand uniquely represents the density $f_Y(y)$ almost everywhere. Therefore:

$$f_Y(y) = \begin{cases} \sum_{k=1}^m f_X(g_k^{-1}(y)) |\det J_{g_k^{-1}}(y)| \mathbb{1}_{g(B_k)}(y) & : y \in \bigcup_{i=1}^m g(B_i) \\ 0 & : \text{otherwise.} \end{cases}$$

□

Part 4b i)

Let $B = \{(x, y) \in \mathbb{R}^2 : x^2 + y^2 < 1\}$ and let $g(x, y) := \left(\sqrt{x^2 + y^2}, \arctan(y/x)\right) = (R(x, y), S(x, y)) := (r, s)$. We partition B into three disjoint sets:

$$B_0 := \{(x, y) \in B : x = 0\}$$

$$B_1 := \{(x, y) \in B \mid x > 0\}$$

$$B_2 := \{(x, y) \in B \mid x < 0\}$$

Note that g is not continuously differentiable at $x = 0$. Also, g is not injective on $B_1 \cup B_2$ because for any $(x, y) \in B_1$:

$$\begin{aligned} g(-x, -y) &= \left(\sqrt{(-x)^2 + (-y)^2}, \arctan(-y/-x)\right) \\ &= \left(\sqrt{x^2 + y^2}, \arctan(y/x)\right) \\ &= g(x, y) \end{aligned}$$

where $(-x, -y) \in B_2$.

We analyze the three sets separately to check that the conditions of part a) are satisfied and to determine the corresponding inverse functions and Jacobian matrices:

- **Analysis of B_0 :** B_0 is a one-dimensional line segment and thus a Lebesgue null set in $\mathbb{R}^2$.
- **Analysis of B_1 :** On B_1 , $x > 0$, $r = \sqrt{x^2 + y^2}$ with $x^2 + y^2 < 1$ implies that $r \in (0, 1)$ and $s = \arctan(y/x) \in (-\pi/2, \pi/2)$, i.e., the image of $g(B_1) = (0, 1) \times (-\pi/2, \pi/2)$. We then proceed to calculate the inverse:

$$s = \arctan(y/x) \implies \tan(s) = y/x \implies y = x \tan(s) \quad (4b.1)$$

$$r = \sqrt{x^2 + y^2} \stackrel{(4b.1)}{=} \sqrt{x^2 + x^2 \tan^2(s)} = |x| \sqrt{1 + \tan^2(s)} = |x| |\sec(s)| \quad (4b.2)$$

Since $x > 0$ on B_1 and $\cos(s) > 0$ for $s \in (-\pi/2, \pi/2)$, we have that $|x| = x$ and $|\sec(s)| = \sec(s) = 1/\cos(s)$, then:

$$r = \frac{x}{\cos(s)} \implies x = r \cos(s)$$

Substituting back into equation (4b.1):

$$y = x \tan(s) = r \cos(s) \cdot \frac{\sin(s)}{\cos(s)} = r \sin(s).$$

Then, the inverse, its Jacobian matrix, and its determinant are given by:

$$g_1^{-1}(r, s) = (r \cos(s), r \sin(s))$$

$$J_{g_1^{-1}}(r, s) = \begin{pmatrix} \frac{\partial x}{\partial r} & \frac{\partial x}{\partial s} \\ \frac{\partial y}{\partial r} & \frac{\partial y}{\partial s} \end{pmatrix} = \begin{pmatrix} \cos(s) & -r \sin(s) \\ \sin(s) & r \cos(s) \end{pmatrix}$$

$$|\det J_{g_1^{-1}}(r, s)| = |r \cos^2(s) + r \sin^2(s)| = r(\cos^2(s) + \sin^2(s)) = r \in (0, 1)$$

Since the determinant is strictly positive, $J_{g_1^{-1}}(r, s)$ is nowhere singular.

- **Analysis of B_2 :** On B_2 , $x < 0$, $r = \sqrt{x^2 + y^2}$ with $x^2 + y^2 < 1$ implies that $r \in (0, 1)$ and $s = \arctan(y/x) \in (-\pi/2, \pi/2)$, i.e., the image of $g(B_2) = (0, 1) \times (-\pi/2, \pi/2)$. We then proceed to calculate the inverse. By 4b.1 and 4b.2, we have $y = x \tan(s)$ and $r = |x| |\sec(s)|$. As before, since $\cos(s) > 0$ for $s \in (-\pi/2, \pi/2)$, we have that $|\sec(s)| = \sec(s) = 1/\cos(s)$. However, since $x < 0$ on B_2 and we have that $|x| = -x$, then:

$$r = \frac{-x}{\cos(s)} \implies x = -r \cos(s),$$

$$y = x \tan(s) = -r \cos(s) \cdot \frac{\sin(s)}{\cos(s)} = -r \sin(s).$$

Then, the inverse, its Jacobian matrix, and its determinant are given by:

$$g_2^{-1}(r, s) = (-r \cos(s), -r \sin(s))$$

$$J_{g_2^{-1}}(r, s) = \begin{pmatrix} \frac{\partial x}{\partial r} & \frac{\partial x}{\partial s} \\ \frac{\partial y}{\partial r} & \frac{\partial y}{\partial s} \end{pmatrix} = \begin{pmatrix} -\cos(s) & r \sin(s) \\ -\sin(s) & -r \cos(s) \end{pmatrix}$$

$$|\det J_{g_2^{-1}}(r, s)| = |r \cos^2(s) + r \sin^2(s)| = r(\cos^2(s) + \sin^2(s)) = r \in (0, 1)$$

Since the determinant is strictly positive, $J_{g_2^{-1}}(r, s)$ is nowhere singular.

For B_1 and B_2 , the joint density of the uniform distribution on the unit disk is $f_{(X,Y)}(x, y) = \frac{1}{\pi}$. Applying the generalized density transformation formula of part a):

$$\begin{aligned} f_{(R,S)}(r, s) &= \sum_{k=1}^2 f_{(X,Y)}(g_k^{-1}(r, s)) |\det J_{g_k^{-1}}(r, s)| \mathbb{1}_{g(B_k)}(r, s) \\ &= \left(\frac{1}{\pi} \cdot r \right) \mathbb{1}_{(0,1) \times (-\pi/2, \pi/2)}(r, s) + \left(\frac{1}{\pi} \cdot r \right) \mathbb{1}_{(0,1) \times (-\pi/2, \pi/2)}(r, s) \\ &= \frac{2r}{\pi} \mathbb{1}_{(0,1) \times (-\pi/2, \pi/2)}(r, s) \end{aligned}$$

Part 4b ii)

By Theorem 1.5.11, the marginal density $f_R(r)$ is obtained by integrating the joint density $f_{(R,S)}(r, s)$ over all possible values of $s \in \mathbb{R}$.

For $r \notin (0, 1)$, $f_R(r) = 0$. For $r \in (0, 1)$, the indicator function restricts the bounds of integration to $(-\pi/2, \pi/2)$:

$$\begin{aligned} f_R(r) &= \int_{-\infty}^{\infty} f_{(R,S)}(r, s) ds \\ &= \int_{-\pi/2}^{\pi/2} \frac{2r}{\pi} ds \\ &= \frac{2r}{\pi} \left[s \right]_{-\pi/2}^{\pi/2} \\ &= \frac{2r}{\pi} \left(\frac{\pi}{2} - \left(-\frac{\pi}{2} \right) \right) \\ &= 2r \end{aligned}$$

Therefore, the marginal density of R is $f_R(r) = 2r \mathbb{1}_{(0,1)}(r)$.